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Overview

The **Workflow Canvas** (also called Flow Designer Canvas) is a visual graph-based workflow builder that lets you design and manage workflow structure using an interactive canvas. Instead of text-based configuration, you can see your entire workflow as connected nodes and edges, making it easy to understand workflow paths and dependencies at a glance.

The canvas uses React Flow with automatic layout powered by the ELK (Eclipse Layout Kernel) algorithm, ensuring your workflow is always displayed with optimal clarity and minimal edge crossings.

flow Canvas showing complete workflow with queues

Key Benefits

- **Visual Design** - See your entire workflow structure in one view
- **Drag and Drop** - No code required to build complex workflows
- **Real-time Layout** - Automatic arrangement keeps workflows organized
- **Multiple Edge Types** - Clearly distinguish between manual actions, rules, and events
- **Context Panels** - Configuration panels appear when you need them
- **Interactive Navigation** - Click on nodes to explore details

Accessing the Canvas

1. Navigate to **Admin > ResolveX > Workflows**
2. Find your workflow in the list
3. Click the **Actions** (...) menu next to the workflow
4. Select **Flow Designer**

The canvas opens showing your workflow structure with all queues, actions, rules, and events.

 Navigation path to access Flow Designer from workflows list

Canvas Layout

The canvas displays your workflow as a left-to-right graph:

- **Workflow Node** (root) - Starting point representing the workflow itself
- **Queue Nodes** - Each stage in your workflow appears as a node with 8 interactive icons
- **Edges** - Connections between queues showing actions, rules, and events
- **Legend** - Visual guide to edge types and colors

Canvas layout showing workflow node

Node Types

Node Type	Appearance	Purpose
Workflow Node	Root node with workflow name	Starting point showing workflow context
Queue Node	Rectangular card with 8 icons	Represents a workflow stage with navigation and action icons

Edge Types

Edges have distinct visual styles so you can quickly identify routing logic:

Edge Type	Visual Style	Icon	Purpose
Action Edge	Solid indigo line	None	Manual user action connecting two queues
Rule Edge	Dashed green line		Condition-based automatic routing
Event Edge	Dashed amber/yellow line		Time or schedule-based automatic actions

Legend showing three edge types with different colors and styles

TIP

Toggle between line and tube edge styles using the edge style button in the canvas toolbar.

Queue Node Icons

Each queue node displays **8 interactive icons** organized in two rows:

Queue node showing all 8 icons in two rows

Navigation Icons (Top Row)

These icons open context-sensitive panels showing queue details:

Icon	Name	Opens
	Overview	Queue summary, description, role configuration, item count
	Manual Actions	List of user-triggered actions that move items from this queue
	Rules	Condition-based rules linked to this queue with priorities and target queues
	Events	Time/schedule-based events linked to this queue with action configurations
	System Actions	Role configuration for system-managed actions (Manage Notes, Manage Attachments)

Action Icons (Bottom Row)

These icons open dialogs to add new routing logic:

Icon	Name	Creates
	Add Manual Action	New user-triggered action with target queue
	Add Rule Link	Link existing rule to this queue with priority and actions
	Add Event Link	Link existing event to this queue with scope and actions

NOTE

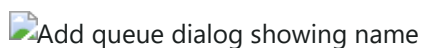
Action icons only appear when you have the necessary permissions to modify workflow configuration.

Working with Queues

Adding a Queue

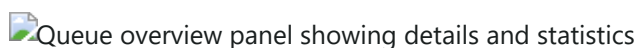
1. Click the **Add Queue** button in the canvas toolbar
2. Enter **Queue Name** (e.g., "Manager Review")
3. Enter **Queue Description**
4. Select **Queue Roles** (who can see items in this queue)
5. Click **Save**

The new queue appears on the canvas with automatic layout.



Viewing Queue Details

1. Click the **Info** icon on any queue node
2. The right panel opens showing:
 - a. Queue name and description
 - b. Assigned roles
 - c. Current item count
 - d. Queue statistics



Configuring Queue Roles

1. Click the **Info** icon on the queue node
2. Click **Edit Roles** in the overview panel
3. Select roles from the dropdown
4. Click **Save**

Users must have queue roles to see items in that queue.

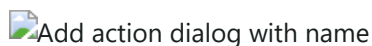
Working with Actions

Actions are manual operations that users perform to move items between queues.

Adding a Manual Action

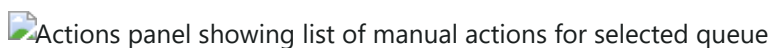
1. Click the **+ Add Action** icon on the source queue
2. Enter **Action Name** (e.g., "Approve", "Escalate")
3. Enter **Action Description**
4. Select **Target Queue** (where items will move)
5. Select **Action Roles** (who can perform this action)
6. Click **Save**

A solid indigo edge appears connecting the source to target queue.



Viewing Manual Actions

1. Click the **Manual Actions** icon on a queue node
2. The panel shows all actions with:
 - a. Action name and description
 - b. Target queue
 - c. Assigned roles
3. Click on any action to edit or delete

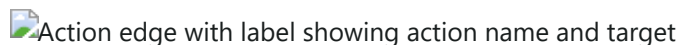


Action Edge Labels

Action edges display labels showing:

- Action name
- Target queue name (if space permits)

Click on an action edge to see full action details.



Working with Rules

Rules are conditions that automatically route items based on FEEL expressions.

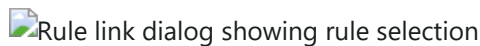
Prerequisite: Create Rules

Before linking rules to workflows, you must create them in **Admin > ResolveX > Rules**. See [Rules Configuration](#) for details on creating rules with FEEL expressions.

Linking a Rule to a Queue

1. Click the **+** **Add Rule Link** icon on the source queue
2. Select **Rule** from the dropdown (shows existing rules)
3. Set **Priority** (lower number = evaluated first)
4. Add **Actions** - typically MOVE_QUEUE to target queue
5. Click **Save**

A dashed green edge with split icon appears showing the rule connection.



Viewing Rules for a Queue

1. Click the **Rules** icon on a queue node
2. The panel shows all linked rules with:
 - a. Rule name and description
 - b. Rule expression (FEEL syntax)
 - c. Priority
 - d. Target queue(s)
3. Click on any rule to edit mapping or delete link



Rule Evaluation Order

Rules are evaluated in priority order when an item enters a queue:

1. Item enters queue
2. Rules evaluated from lowest priority number to highest (1, 2, 3...)
3. First matching rule executes → item moves
4. Remaining rules skipped (**first match wins**)

TIP

Set more specific rules to lower priority numbers (higher priority) so they're checked first.

Working with Events

Events are time or schedule-based triggers that automatically execute actions on workflow items.

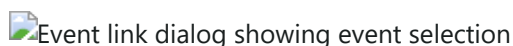
Prerequisite: Create Events

Before linking events to workflows, you must create them in **Admin > ResolveX > Events**. See [Events Configuration](#) for details on creating CRON and TIME events.

Linking an Event to a Queue

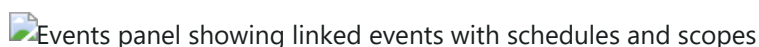
1. Click the **+ +**  **Add Event Link** icon on the target queue
2. Select **Event** from the dropdown (shows existing events)
3. Choose **Action Scope**:
 - a. **ITEM** - Execute actions per-item individually
 - b. **BATCH** - Execute actions on all items at once
4. (Optional) Select **Filter Rule** to limit which items get actions
5. (For **BATCH**) Set **Max Batch Size** and payload type
6. Add **Actions** (MOVE_QUEUE, SEND_EMAIL, etc.)
7. Click **Save**

A dashed amber edge with bell icon appears showing the event connection.



Viewing Events for a Queue

1. Click the  **Events** icon on a queue node
2. The panel shows all linked events with:
 - a. Event name and type (CRON/TIME)
 - b. Event schedule or duration
 - c. Action scope (ITEM/BATCH)
 - d. Filter rule (if any)
 - e. Actions to execute
3. Click on any event to edit mapping or delete link



Event Types on Canvas

Event Type	Canvas Display	Behavior
TIME Event	Amber edge with duration label	Fires once per item after duration passes
CRON Event	Amber edge with schedule label	Fires on recurring schedule for all items
With Filter Rule	Shows filter rule name in edge label	Only matching items receive actions

Working with System Actions

System actions are operations that the platform manages automatically for every queue. Unlike manual actions, they cannot be created, renamed, or deleted — only the roles permitted to use them can be configured.

The two system actions available on every queue are:

System Action	Purpose
Manage Notes	Controls which roles can add and delete notes on work items in this queue
Manage Attachments	Controls which roles can upload and delete file attachments on work items in this queue

NOTE

If neither system action appears in the panel, they have not yet been initialised for this queue. Re-saving the queue triggers automatic creation.

Configuring System Action Roles

1. Click the  **System Actions** icon on the queue node
2. The **System Actions** panel opens on the right
3. Select the action to configure from the **System Action** dropdown:
 - a. **Manage Notes**
 - b. **Manage Attachments**
4. A read-only info card shows the action name and a **System Managed** badge confirming it cannot be edited
5. In the **Action Roles** section, toggle the roles that should be permitted to perform this action

Only roles already assigned to the queue are available for selection.

6. Click **Save Action Roles**

Repeat for each system action.

 System Actions panel showing action selector and role configuration

TIP

Use **Manage Attachments** roles to restrict evidence uploads to senior reviewers, while allowing all queue roles to use **Manage Notes** for general commentary.

Canvas Controls

The canvas toolbar provides controls for navigation and display:

 Canvas toolbar showing all control buttons

Available Controls

Control	Icon	Function
Workflow Selector	Dropdown	Switch between workflows without leaving canvas
Add Queue	+	Create new queue in current workflow

Control	Icon	Function
Zoom In/Out	+/-	Adjust canvas zoom level
Fit View		Auto-zoom to fit entire workflow in viewport
Auto Layout		Re-run ELK layout algorithm to reorganize nodes
Edge Style	Line/Tube toggle	Switch between line and tube edge rendering
Legend		Show/hide edge type legend

Zoom and Pan

- **Zoom:** Use mouse wheel or +/- buttons
- **Pan:** Click and drag on empty canvas space
- **Reset View:** Click **Fit View** to center and scale workflow

TIP

For large workflows, use **Fit View** first, then zoom in to specific sections.

Auto-Layout

The auto-layout feature uses the ELK (Eclipse Layout Kernel) algorithm to arrange nodes with:

- Minimal edge crossings
- Consistent spacing
- Left-to-right flow
- Handle-based edge routing for parallel edges

Click the **Auto Layout** button if nodes appear overlapping or difficult to read.

Comparison showing workflow before and after auto-layout

Legend Panel

The legend panel explains edge types and visual conventions:

Legend panel showing edge types

The legend shows:

- **Action Edge** - Solid indigo line (manual user actions)
- **Rule Edge** - Dashed green line with split icon (condition-based routing)
- **Event Edge** - Dashed amber line with bell icon (time/schedule triggers)

Click the legend button in the toolbar to toggle visibility.

Best Practices

Workflow Organization

- **Keep workflows focused** - One workflow per business process (e.g., "Payment Disputes", "Loan Approvals")
- **Use descriptive queue names** - "Manager Review" is better than "Queue 2"
- **Limit queue count** - More than 10-12 queues becomes hard to visualize
- **Group related queues** - Use naming conventions like "Review - L1", "Review - L2"

Edge Routing

- **Minimize edge crossings** - Use auto-layout to optimize node placement
- **Use clear action names** - Edge labels should explain the operation
- **Distinguish edge types** - Use rules for conditional logic, events for time-based actions
- **Avoid circular flows** - Items should progress through workflow, not loop indefinitely

Rules and Events Strategy

- **Rules for conditional routing** - Use when routing depends on item data
- **Events for SLA enforcement** - Use TIME events to escalate overdue items
- **Events for reminders** - Use CRON events for daily/weekly notifications
- **Combine rules with events** - Use filter rules with CRON events for targeted processing

Performance Considerations

- **Limit rule priority conflicts** - Ensure each rule has unique priority
- **Test rules before production** - Use test workflows to validate rule logic
- **Monitor event execution** - Check scheduler logs to confirm events fire as expected
- **Use batch scope wisely** - BATCH events process faster but less granular control

Common Workflows

Simple Approval Workflow

```
New Cases → [Approve] → Closed  
           → [Reject] → Rejected
```

TEXT

Two manual actions from single queue.

Multi-Stage Review Workflow

```
New Cases → [Assign] → L1 Review → [Approve] → L2 Review → [Approve] → Closed  
                                   → [Reject] → Rejected
```

TEXT

Three queues with progressive approval stages.

Rule-Based Routing Workflow

New Cases → [Rule: High Value] → Manager Review → [Approve] → Closed
→ [Rule: Standard Value] → Auto Approved → Closed
→ [Rule: Low Value] → Auto Rejected → Rejected

TEXT

Rules automatically route based on item amount.

Event-Driven SLA Workflow

Pending → [Action: Resolve] → Closed
→ [Event: PT2H] → Escalated → [Action: Resolve] → Closed

TEXT

TIME event escalates items sitting in Pending > 2 hours.

Troubleshooting

See [Canvas Issues](#) for common problems and solutions.

Related Topics

- [Workflow Configuration](#)
- [Rules Configuration](#)
- [Events Configuration](#)
- [Workflow Design Best Practices](#)
- [Terminology Reference](#)



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